



Islamic University of Technology

Semester: 2ND

Course: PHY4241

Assignment No: 02

Assignment Title: CHAPTER - 03.

Submitted By:

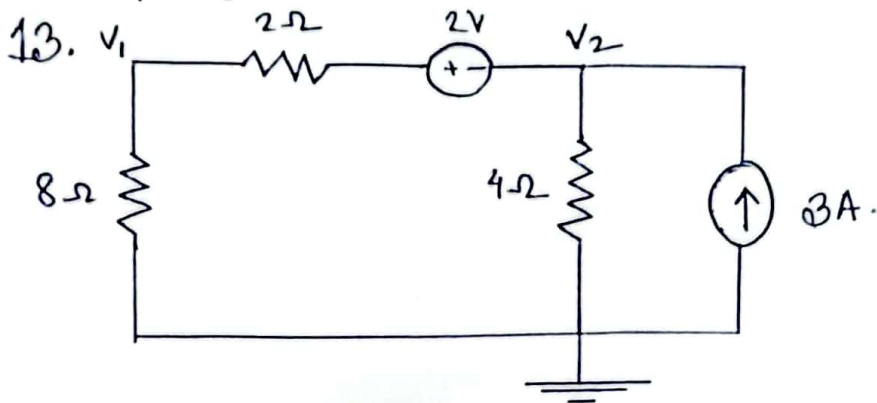
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Section:01

Dept. of Computer Science and Engineering (CSE)

Chapter-03:



Q. Calculate v_1 and v_2 using nodal analysis.

Solution:

At node 2:

$$\frac{v_2 + 2}{10} + \frac{v_2}{4} = 3.$$

$$\Rightarrow 4v_2 + 8 + 10v_2 - 120 = 0$$

$$\Rightarrow v_2 = \frac{112}{14}$$

$$\therefore v_2 = 8 \text{ volts.}$$

From the circuit:

$$I = \frac{v_2 + 2 - 0}{10}$$

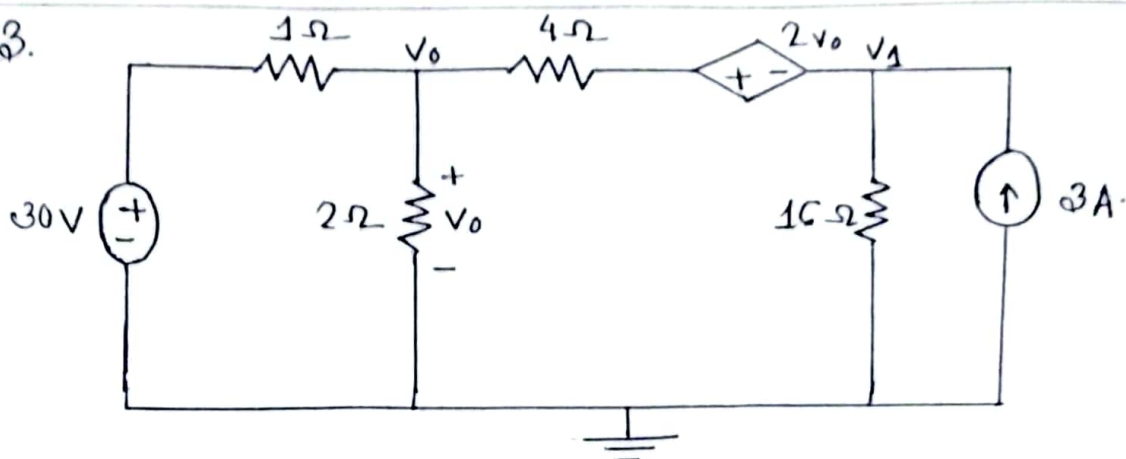
$$= \frac{8 + 2}{10}$$

$$= 1 \text{ A.}$$

$$\therefore v_1 = 8 \times 1 = 8 \text{ volts}$$

Ans

23.



Q. Use nodal analysis to find V_0 in the circuit.

Solution: At node-0,

$$\frac{v_0 - 30}{1} + \frac{v_0 - 0}{2} + \frac{v_0 - (2v_0 + v_1)}{4} = 0$$

$$\Rightarrow 4v_0 - 120 + 2v_0 + v_0 - 2v_0 - v_1 = 0$$

$$\Rightarrow 5v_0 - v_1 = 120 \quad \text{--- (i)}$$

At node-1,

$$\frac{(2v_0 + v_1) - v_0}{4} + \frac{v_1}{16} = 3.$$

$$\Rightarrow 8v_0 + 4v_1 - 4v_0 + v_1 - 48 = 0$$

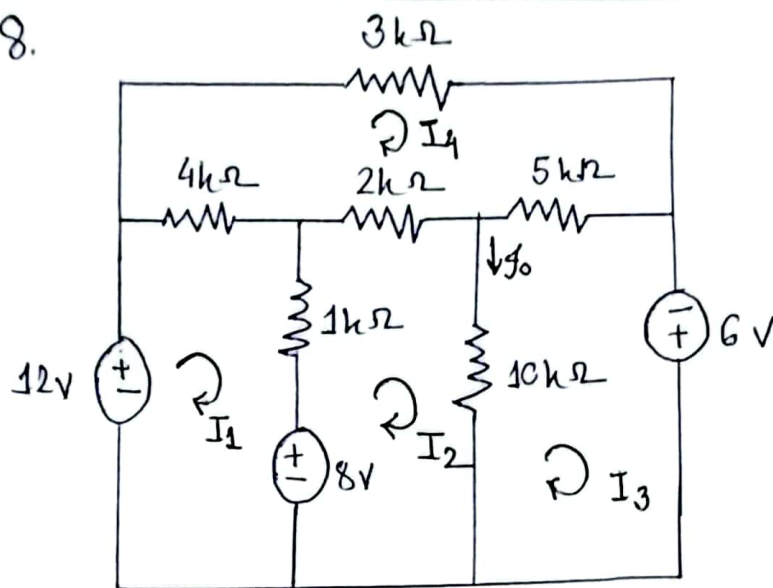
$$\Rightarrow 4v_0 + 5v_1 = 48 \quad \text{--- (ii)}$$

Solving (i) and (ii), we get:

$$v_0 = 22.34 \text{ volts.}$$

Ans.

48.



Determine the current through $10k\Omega$ resistor using mesh analysis.

Solution: We consider the current flow to be in mA. units.

For mesh 01,

$$-12 + 4(I_1 - I_4) + 1(I_1 - I_2) + 8 = 0$$

$$\Rightarrow 5I_1 - I_2 - 4I_4 = 4. \quad \text{--- (i)}$$

For mesh 02,

$$-8 + I_2 - I_1 + 2(I_2 - I_4) + 10(I_2 - I_3) = 0$$

$$\Rightarrow -I_1 + 13I_2 - 10I_3 - 2I_4 = 8 \quad \text{--- (ii)}$$

For mesh 03,

$$10(I_3 - I_2) + 5(I_3 - I_4) - 6 = 0$$

$$\Rightarrow -10I_2 + 15I_3 - 5I_4 = 6 \quad \text{--- (iii)}$$

For mesh 04,

$$4(I_4 - I_1) + 2(I_4 - I_2) + 5(I_4 - I_3) + 3I_4 = 0$$
$$\Rightarrow -4I_1 - 2I_2 - 5I_3 + 14I_4 = 0 \quad \text{--- (iv)}$$

Solving the equations, we get:

$$I_1 = 7.218 \text{ A}$$

$$I_2 = 8.08 \text{ A}$$

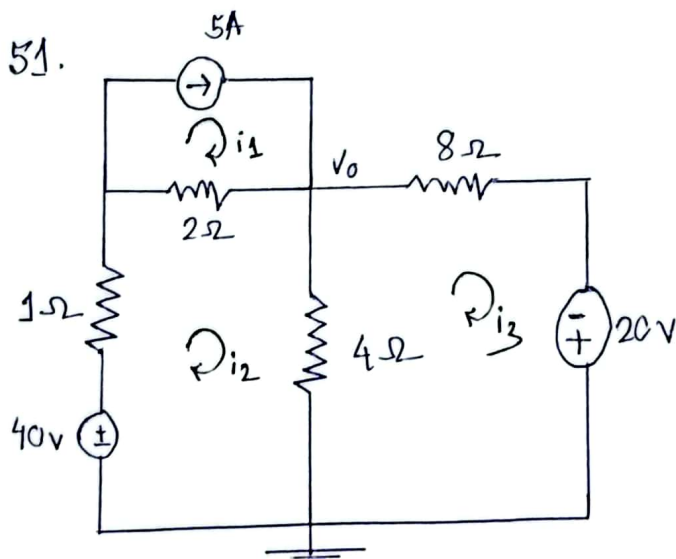
$$I_3 = 7.791 \text{ A}$$

$$I_4 = 6 \text{ A}$$

From the circuit, $I_o = I_2 - I_3$

$$= (8.08 - 7.791) \text{ mA}$$
$$= 0.289 \text{ mA}$$

Ans.



Apply mesh analysis to find v_o in the circuit.

For mesh 01,

$$i_1 = 5$$

For mesh 02,

$$-40 + i_2 + 2i_2 - 2i_1 + 4i_2 - 4i_3 = 0$$

$$\Rightarrow -2i_1 + 7i_2 - 4i_3 = 40$$

$$\Rightarrow -2(5) + 7i_2 - 4i_3 = 40$$

$$\Rightarrow 7i_2 - 4i_3 = 50 \quad \text{--- (i)}$$

For mesh 03,

$$4i_3 - 4i_2 + 8i_3 - 20 = 0$$

$$\Rightarrow -4i_2 + 12i_3 = 20 \quad \text{--- (ii)}$$

Solving (i) and (ii), we get:

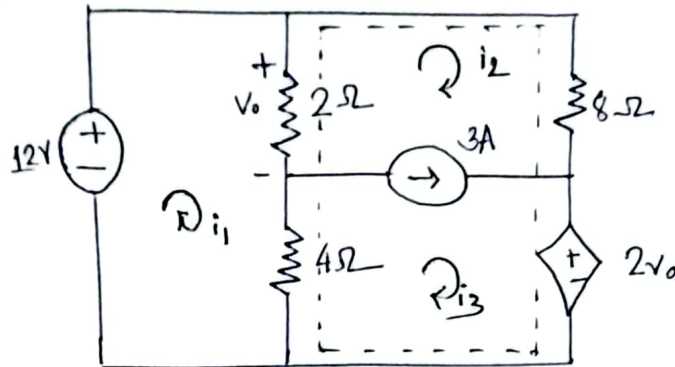
$$i_2 = 10A, \quad i_3 = 5A.$$

$$\begin{aligned} \therefore V_o &= 4(i_2 - i_3) \\ &= 4(10 - 5) \\ &= 20 \text{ volts} \end{aligned}$$

Ans.



52.



Use mesh analysis to find i_1, i_2, i_3 in the circuit.

Solution:

For mesh 01,

$$2(i_1 - i_2) + 4(i_1 - i_3) - 12 = 0$$

$$\Rightarrow 6i_1 - 2i_2 - 4i_3 = 12$$

$$\Rightarrow 3i_1 - i_2 - 2i_3 = 6 \quad \text{--- (i)}$$

For the supermesh,

$$2i_2 - 2i_1 + 8i_2 + 2v_o + 4i_3 - 4i_1 = 0$$

$$\Rightarrow 2i_2 - 2i_1 + 8i_2 + 2\{2(i_2 - i_1)\} + 4i_3 - 4i_1 = 0$$

$$[\because v_o = 2(i_2 - i_1)]$$

$$\Rightarrow -i_1 + 3i_2 + 2i_3 = 0 \quad \text{--- (ii)}$$

For the independent current source,

$$i_3 = 3 + i_2 \quad \text{--- (iii)}$$

Solving the equations, we get,

$$i_1 = 3.3A, \quad i_2 = -0.55A, \quad i_3 = 2.45A \quad \text{Ans.}$$